

TIA–FPGA – FPGA replacement for the Atari 2600 Television Interface Adaptor (C010444)

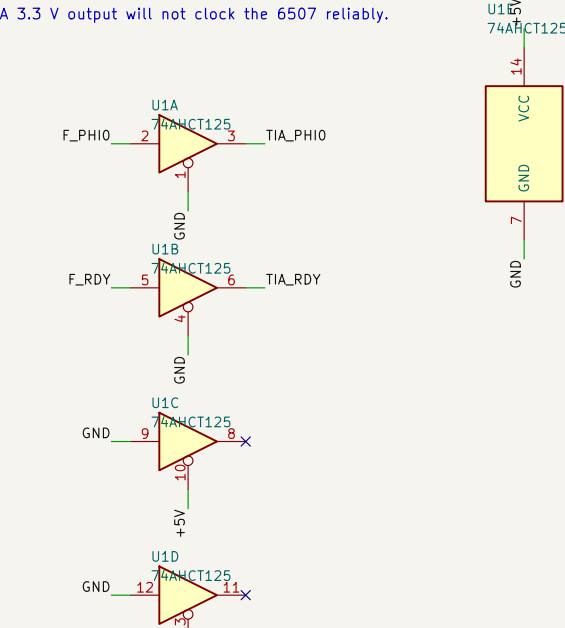
Rev A – digital interface only. Composite video and paddles are deferred to later revisions.

FPGA MODULE – Sipeed Tang Nano 9K (GW1NR–LV9QN88PC6/I5)

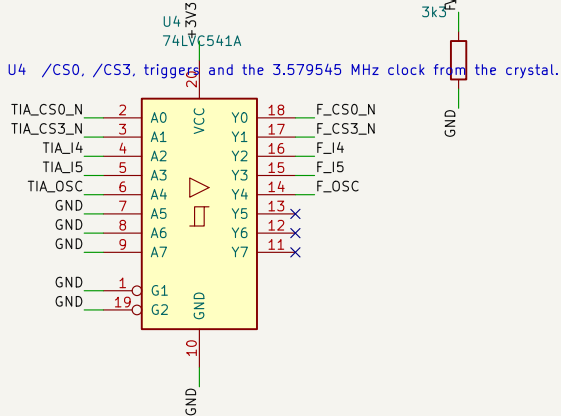
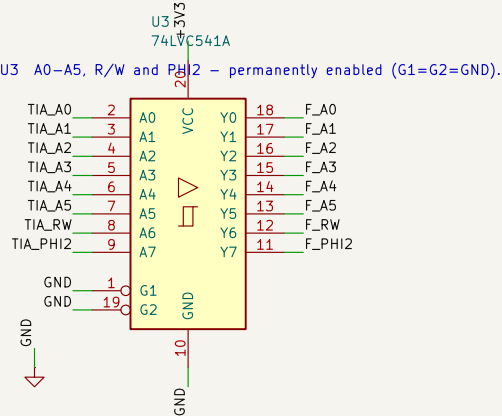
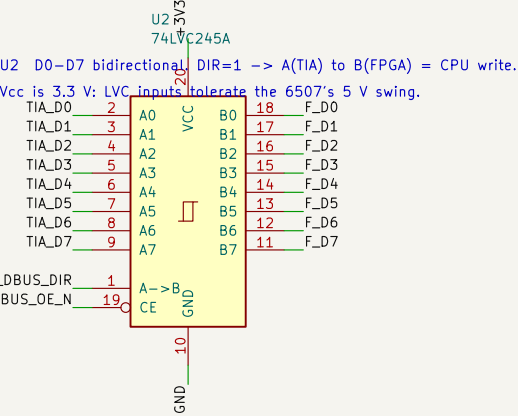
Header numbering follows the Sipeed silkscreen.
FPGA pin assignment lives in fpga/tia-fpga.cst – see the TODOs there.

LEVEL TRANSLATORS

U1 is the critical part: the NMOS 6507 requires VIH(PHI0) = Vcc–0.2 V ~ 4.8 V.
A 3.3 V output will not clock the 6507 reliably.

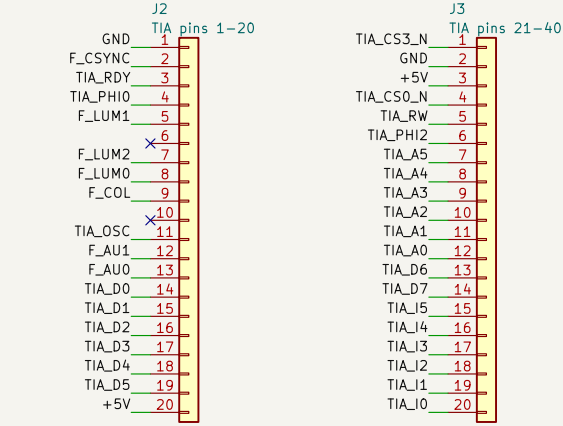


U1 runs from +5 V – this is where the level step–up happens.
Spare buffers 3 and 4: /OE tied to +5 V, input tied to GND.



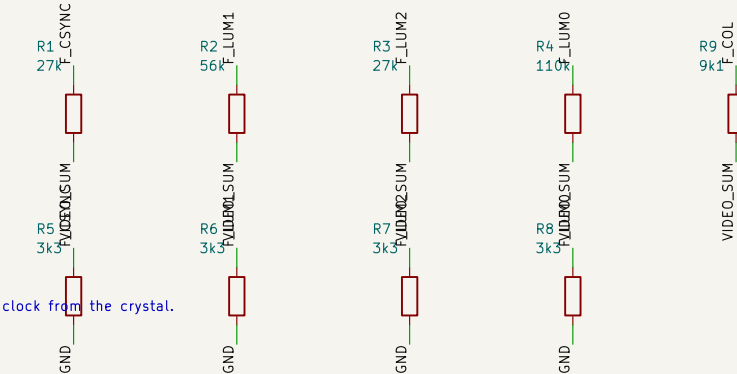
TIA SOCKET – DIP–40

Two 20–way SIP strips that plug into the 2600 socket
or a breadboard, in place of the C010444.



VIDEO OUTPUT – CX–2600A resistor network (phase 6, not populated)

4:2:1 luma weighting. Output stage (2N3563) follows the original sheet.



POWER AND DECOUPLING

+3V3 comes from the Tang Nano itself. +5 V comes from the breadboard rail.

